

DEVELOPMENT OF LATERAL SURFACES

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[PRISM PROBLEMS]

[Cutting plane passing through the vertical edges**]**

[1] A **square prism** of **base side 40mm** and **axis length 65mm** is **resting on HP** on its base with all the vertical faces being equally inclined to VP. It is cut by a plane inclined at **40° to HP**, perpendicular to VP cuts **the axis at its mid height**. Draw the development of the lower portion of the prism.

[2] A **rectangular prism** of base **40mm × 25mm** and **height 65mm** rests **on HP** on its base with the longer base side inclined at 30° to VP. It is cut by a plane inclined at 40° to HP, perpendicular to VP cuts the axis at its mid height. Draw the development of the remaining portion of the prism. **[Sketch book problem]**

[3] A regular pentagonal prism of height 60mm and base edges 30mm rests with its **base on HP**. The vertical face closest to VP is 30° to it. Draw the development of the truncated prism with its truncated surface inclined at 60° to the axis and bisecting it. [Sketch book problem]

[PRISM PROBLEMS]

[Cutting plane passing through the vertical, top & bottom edges**]**

[4] A square prism of **base side 40mm** and **axis length 65mm** is resting on HP on its base with all the vertical faces being equally inclined to VP. It is cut by a plane inclined at 60° to HP, perpendicular to VP cuts the axis at a distance 55mm from the base. Draw the development of the lower portion of the prism. [Sketch book problem]

[5] A rectangular prism of base 40mm × 25mm and height 65mm **rests on HP** on its base with the longer base side inclined at 30° to VP. It is cut by a plane inclined at 60° to HP and perpendicular to VP and passes through the extreme left corner of the base. Draw the development of the remaining portion of the prism. [Sketch book problem]

[6] A regular pentagonal prism of **height 60mm** and **base edges 30mm** rests with its base on HP. The vertical face closest to VP is 30° to it. Draw the development of the truncated prism with its truncated surface inclined at 20° to the axis so as to pass through a point on it 40mm above the base. Develop the truncated portion of the prism so as to produce a one-piece development. [Sketch book problem]

[7] Draw the development of the truncated portion of the lateral faces of a pentagonal prism of 20mm sides of base and 50mm height standing vertically with one of its rectangular faces parallel to VP and nearer to it so as to produce one-piece development. The inclined face of the truncated prism is 30° to its axis and passes through the right extreme corner of the top face of the prism.

[8] A square prism of base side 40mm and axis length 65mm is resting on HP on its base with all the vertical faces being equally inclined to VP. It is cut by an inclined plane 60° to HP and perpendicular to VP and is passing through a point on the axis at a distance 15mm from the top face. Draw the development of the lower portion of the prism.

[9] A cube of side 40mm is resting on HP with its base on HP such that one of its vertical faces is inclined at 30° to VP. It is cut by a section plane perpendicular to VP, inclined to HP at an angle 45° and passes through the mid-point of the axis. Draw the development of the lower lateral surfaces of the cube.[Sketch book problem]

[PYRAMID PROBLEMS]

[10] A **square pyramid** of side of base 45mm, **altitude 70mm** is resting with its base on HP with two sides of base parallel to VP. The pyramid is cut by a section plane, which is perpendicular to the VP and inclined at 40° to the HP. The cutting plane bisects the axis of the pyramid. Obtain the development of the lateral surface of truncated pyramid. [Sketch book problem]

[11] A square pyramid **base 40mm side** and **axis 65mm long** has its base on HP and all the edges of the base are equally inclined to VP. It is cut to with an inclined section plane so as the truncated surface at 45° to its axis, bisecting it. Draw the development of the truncated pyramid.

[12] A **frustum of a square pyramid** has its base 40mm sides and top 16mm sides, total height of a square pyramid is 60mm, its axis is vertical and a side of its base is parallel to VP. Draw the projections of the frustum and show the development of the lateral surfaces of it. [Sketch book problem]

[13] A regular pentagonal pyramid of side of base 35mm and altitude 65mm has its base on HP with a side of base perpendicular to VP. The pyramid is cut by a section plane, which is perpendicular to the VP and inclined at 30° to HP. The cutting plane meets the axis of the pyramid at a point 30mm below the vertex. Obtain the development of the remaining part of the pyramid. [Sketch book problem]

[14] A hexagonal pyramid, base sides 25mm and height 60mm, is resting with its base on HP and an edge of base inclined at 40° to VP. It is cut to the shape of the truncated pyramid with the truncated surface indicated in the front view at a point on the axis 20mm from the apex and inclined at 40° to XY. Draw the projections and show the development of the lateral surface of the remaining portion of the pyramid.

[15] A hexagonal pyramid of sides 35mm and altitude 65mm is resting on HP on its base with two of the base sides perpendicular to VP. The pyramid is cut by a plane inclined at 30° to HP and perpendicular to VP and is intersecting the axis at 30mm above the base. Draw the development of the remaining portion of the pyramid. [Sketch book problem]

[CYLINDER PROBLEMS]

[16] A vertical cylinder of base diameter 45mm and axis length 60mm is cut by a plane perpendicular to VP and inclined at 50° to HP, is passing through the centre point of the top face. Draw the development of the lateral surface of the cylinder.

[17] Draw the development of the lateral surface of a truncated vertical cylinder. 40mm diameter of base and height 50mm, the truncated flat surface of the cylinder bisects the axis at 60° to it. [Sketch book problem]

[CONE PROBLEMS]

[18] A right cone of 55mm diameter of base and 75mm height stands on its base on HP. It is cut to the shape of a truncated cone with its truncated surface inclined at 45° to the axis lying at a distance of 40mm from the apex of the cone. Obtain the development of lateral surface of the truncated cone. [Sketch book problem]

$$\theta = \frac{360 \times r}{L} \rightarrow \text{degrees}$$